



Republic of the Philippines
North Eastern Mindanao State University

OUTCOMES-BASED COURSE SYLLABUS IN
CS 328 – MACHINE LEARNING
2nd Semester A.Y. 2025-2026

PHILOSOPHY

NEMSU believes that higher education is an instrument for the improvement of life through democratized access to quality education in the development of a well-rounded person.

VISION

A Research University advancing technology and innovation for sustainable development.

MISSION

We drive sustainable development through quality instruction, innovate research, community collaboration, and technological advancement.

CORE VALUES

COMPASSION

Compassion entails our value of promoting empathy, sincerity, and authenticity within the academic community to foster a caring and supportive environment where individuals feel understood, valued, and respected.

ACCOUNTABILITY

Accountability signifies our unwavering commitment to integrity, honesty, and transparent practices, coupled with a sense of responsibility for our actions and decisions, ensuring ethical, efficient, and cost-effective stewardship of resources for the common good.

RESPONSIVENESS

Responsiveness is a prompt action and release consistent communication that is focused on providing correct and complete action and/or information to clients and stakeholders.

EXCELLENCE

Excellence means our consistent pursuit of the highest standards of performance, characterized by innovation, dedication, and impactful contributions across all aspects of endeavor.

SERVICE

Service is the embodiment of our professionalism, dedication, and a service-oriented mindset committed to fulfilling our mission with excellence, integrity, and continual improvement, while fostering interdependence, collaboration, and sustainable success within the community and nation-building endeavors.

NEMSU CARES...

These core values transcend mere descriptions of our tasks or the strategies we employ; they form the foundational principles guiding our actions and interactions. As we internalize our responsibilities to realize our University's mission and vision, they serve as the fundamental elements shaping our approach to work and our relationships with clients and stakeholders. Moreover, they play a pivotal role in molding our students into competent, innovative, globally competitive, and service-oriented individuals.

QUALITY POLICY

The North Eastern Mindanao State University commits itself to produce highly motivated, globally competitive, and morally upright human resource through the delivery of transformative and quality higher education that conforms to international standards driven by excellent instruction, relevant researches, sustainable extension, and production services. Together with our stakeholders, we shall endeavor for continual improvement of our quality management system in consonance with statutory and regulatory requirements for clients and industry satisfaction for quality of life.

INSTITUTIONAL OUTCOMES (IOs)

- IO 1.** Imbue transformative education through global competency and innovation for a future-ready workforce.
- IO 2.** Acculturate the values of CARES (Compassion, Accountability, Responsiveness, Excellence, and Service) in the workplace and the community through indigenous knowledge and traditions for sustainable development.
- IO 3.** Propagate the culture of industry-driven research, need-based extension, and entrepreneurial production for sustainable economic viability, environmental stewardship, gender-bias free, gender equality, and social equity in North Eastern Mindanao and beyond.

GOALS OF THE COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

The College of Information Technology Education (CITE) with its competent faculty, ensures that graduates are equipped to join the national and international workforce imbued with ethical values and ideals in humane environment supportive to national thrust for social transformation and responsibilities in the selection, development, application, integration and management of computing technologies.

The College is committed to:

- 5.1. Ensure high standard in instruction to produce quality and competent graduates;
- 5.2. Guarantee involvement faculty, staff and students in enhancement and training programs to be able to come up with environment-friendly and practical innovation that shall aid in the transfer of technology for community outreach activities;
- 5.3. Strengthen sense of professionalism and ethical values to maintain morals of the society;
- 5.4. Conduct relevant researches and experiments that shall add to existing body of scientific knowledge;
- 5.5. Promote linkages with other academic units and industries, locally, and globally to keep pace with the fast and constantly evolving technology.

PROGRAM GOALS OF THE BACHELOR OF SCIENCE IN COMPUTER SCIENCE

The Bachelor of Science in Computer Science (BSCS) is intended to produce graduates who are globally competent, innovative, and socially and ethically responsible computing professionals. The key program goals generally emphasize:

- ✚ Developing graduates who can apply computing knowledge effectively in various professional and organizational contexts.
- ✚ Encouraging innovation and creativity in solving computing and information technology challenges.
- ✚ Instilling social and ethical responsibility in the practice of computing professions.
- ✚ Preparing graduates to be adaptive, responsive to technological advances, and capable of lifelong learning.

These program goals align BSCS education with both local and global industry standards to ensure graduates are competitive and prepared to contribute meaningfully to society and the computing profession. (CMO No. 25, s. 2015)

PROGRAM DESCRIPTION (BSCS)

The BS Computer Science (BSCS) program includes the study of computing concepts and theories, algorithmic foundations and new developments in computing. The Program prepares students to design and create algorithmically complex software and develop new and effective algorithms for solving computing problems.

The program also includes the study of the standards and practices in Software Engineering. It prepares students to acquire skills and disciplines required for designing, writing and modifying software components, modules and applications that comprise software solutions.

PROGRAM OUTCOMES (POs) <i>Bachelor of Science in Computer Science (BSCS)</i>	COURSE OUTCOMES (COs) <i>(depends per subject; has to align with POs)</i>
1. Apply knowledge of computing fundamentals, knowledge of computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.	CO1. Explain core machine learning concepts and learning paradigms
2. An ability to apply mathematical foundations, algorithmic principles and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehensions of the tradeoffs involved in design choices.	CO2. Apply supervised algorithms to solve machine learning based problems
3. Identify, analyze, formulate, research literature, and solve complex computing problems and requirements reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.	CO3. Apply unsupervised learning algorithms to solve machine learning based problems
4. Create, select, adapt and apply appropriate techniques, resources and modern computing tools to complex computing activities, with an understanding of the limitations to accomplish a common goal.	CO4. Evaluate ML models using appropriate performance metrics
5. Apply computing knowledge with transcultural and international awareness for sustainable community extension development.	CO5. Use ensemble modeling to overcome the limitation of standalone model and improve the overall performance.

COURSE CODE : **CS 328**
COURSE DESCRIPTIVE TITLE : **Machine Learning**
COURSE PRE-REQUISITE : **CS 122, GE-STAT**
CREDIT UNITS : **3 units**
NO. OF CONTACT HOURS : **2 hours lecture and 3 hours laboratory per week**

PERFORMANCE TARGET : 60% of the students should be able to perform the course activities with a Satisfactory to Very Satisfactory performance.

GAD THEMES INTEGRATED : Gender Sensitivity and Awareness - Students will develop console programs that promote inclusiveness by understanding gender concepts, stereotypes and biases.

COURSE DESCRIPTION : The course teaches machine learning fundamentals and provides a solid conceptual foundation. Students learn, use, and critically assess machine learning algorithms through methodical study and experimentation. Students engage with real-world datasets to build practical skills. This allows individuals apply classroom knowledge to real-world issues and construct and analyze models.

DETAILED COURSE LEARNING PLAN

TIME FRAME	TOPICS	LEARNING OUTCOMES (LOs) <i>(At the end of the topic, the student shall be able to:)</i>	PERFORMANCE INDICATORS	INSTRUCTIONAL METHODOLOGY (Online & Face to Face Classes)	LEARNING MATERIALS & RESOURCES	ASSESSMENT / ACTIVITIES
Week 1	NEMSU Vision NEMSU Mission Core Values Quality Policy NEMSU Hymn	Familiarize the University Vision, Mission, Core Values, Quality Policy and Hymn	Comprehension of NEMSU Vision, Mission, Core Values, Quality Policy and NEMSU Hymn	Lecture Class Discussion Round Table Talk	NEMSU Website NEMSU Hymn Video Clip	Oral Recitation
	Institutional Outcomes Program Outcomes Course Outcomes Course Syllabus Course References Course Requirements Course Policies NEMSU GS Grading System	Be oriented with the institutional outcomes (IOs), program outcomes (POs), course outcomes (COs), course syllabus, course references, course requirements, course policies and grading system.	Ability to translate University policies and guidelines into practice	Lecture Interactive Class Discussion	Course Syllabus	Compliance with requirements
Week 2 - 3	🚦 Introduction: ➤ Introduction to	Differentiate classical learning and machine learning	Ability to define ML, its types and	Lecture, discussion, case studies	Reference books,	Quiz, Assignment

	Machine Learning, ➤ Human learning, ➤ ML vs AI vs DL, Types of learning: • Supervised • Unsupervised • Semi-supervised • Reinforcement Learning ➤ Applications of ML, Issues in ML	• Explain the concept, scope, and applications of ML • Differentiate AI, ML and DL • Differentiate various types of ML • Understand use cases of ML • Understand issues of ML	identify real-world use cases, • Demonstration of domain knowledge of AI, ML and DL, • Demonstration of problems from stated ML types	examples, problem-based learning	• PPTs, • online videos, • MOOCs	
Week 4-5	 Model Preparation: ➤ Types of data, ➤ Structure of data, ➤ Data quality and remediation, ➤ Data pre processing	• Understand various types of data • Apply various data preprocessing techniques	Development of programs for data preprocessing	Lecture, Hands-on	• Reference books, • slides, • Jupiter Notebook, • Spider	Assignment, Lab evaluation
Week 6	 Modeling and evaluation: ➤ Predictive and descriptive model, ➤ Training a model, ➤ Model representation and interpretability, ➤ Evaluating performance of model	• Differentiate predictive and descriptive model • Understand flow of model training • Evaluate ML models using metrics	• Use of predictive vs descriptive model • Correct use of performance measures	Lecture, Hands-on	• Reference books • Scikit-learn docs	Quiz, Lab evaluation
Week 7	 Basics of Feature Engineering: ➤ Feature transformation,	• Understanding importance of features • Understand process of feature transformation	• Feature selection and reduction • Variance preservation	Lecture, Hands-on	• Reference books • Research articles	Quiz, Assignment

	➤ Feature subset selection,	• Apply feature selection • Apply PCA for feature reduction	analysis			
Week 8 - 9	 Regression: ➤ Simple Linear Regression, ➤ Multiple Linear Regression, ➤ Assumptions and problems in regression analysis	• Implement linear regression models • Apply advanced regression techniques	• Accurate prediction and error evaluation • Improved model performance metrics	Lecture, Hands-on	• Reference books , • Scikit-learn documentation • Python, • datasets, • Jupiter notebook	Lab evaluation
Week 10 - 11	 Classification: ➤ Classification models and learning steps, ➤ k-Nearest Neighbors, ➤ Logistic Regression, ➤ Decision Trees, ➤ Support Vector Machines	• Implement distance-based classifiers • Construct logistic regression model • Design decision tree model • Apply SVM for classification	• Correct classification and confusion matrix • Proper selection of k-value • Interpretation of decision rules • Proper kernel selection	Lecture, Hands-on	• Reference books , • Scikit-learn documentation • Python, • datasets, • Jupiter notebook	Lab evaluation
Week 12	➤ Midterm Project Submission	Apply fundamental concepts in a mini project.	Development of working console program.	Project Presentation	• Scikit-learn documentation • Python, • datasets, • Jupiter notebook	Project Output Submission
Week 13	MIDTERM EXAM					
Week 14 – 15	 Unsupervised Learning: ➤ Supervised vs	• Differentiate supervised and unsupervised learning • Perform clustering analysis	Creation of programs explaining use of k-Means,	Lecture, Hands-on	• Reference books , • Scikit-learn	Lab evaluation

	unsupervised learning, Clustering: • Partitioning methods, • Hierarchical clustering, • Density based clustering		hierarchical and density based clustering		documentation • Python, • datasets, • Jupiter notebook	
Week 16	🚦 Ensemble learning: • Bagging, • Boosting, • Random Forest	Explain ensemble learning concepts	Improved accuracy and reduced overfitting	Lecture, Hands-on	• Reference books • Research papers, tools	Lab evaluation
Week 17	➤ Final Project Submission	Present and debug final programming project.	Demonstration of functional and well-documented program.	Group/Individual Presentation with panel	• Jupiter Notebook, • Spider	Final Project Submission
Week 18	FINAL EXAM					

COURSE REFERENCES AND SUPPLEMENTAL READINGS

1. Saikat Dutt, S. Chandramouli, A. K. Das. Machine Learnin. Pearson
2. Kevin P. Murphy. Machine Learning: A Probabilistic Perspective, The MIT Press.
3. Tom M. Mitchell. Machine Learning. McGraw-Hill.
4. Andreas Müller & Sarah Guido. Introduction to Machine Learning with Python, O'Reilly.
5. Sebastian Raschka & Vahid Mirjalili. Python Machine Learning, Packt
6. Christopher M. Bishop. Pattern Recognition and Machine Learning, Springer.

WEB RESOURCES:

1. <https://www.coursera.org/specializations/machine-learning-introduction>
2. <https://www.coursera.org/learn/machine-learning-with-python>
3. <https://www.coursera.org/professional-certificates/ibm-machine-learning>
4. <https://www.mygreatlearning.com/academy/premium/machine-learning-essentials-with-python>
5. <https://www.mygreatlearning.com/academy/learn-for-free/courses/python-for-machine-learning>
6. https://onlinecourses.nptel.ac.in/noc26_cs74/preview
7. <https://machinelearningmastery.com/blog/>
8. <https://www.tableau.com/learn/articles/blogs-about-machine-learning-artificial-intelligence>
9. <https://www.mygreatlearning.com/blog/what-is-machine-learning/>
10. <https://www.kaggle.com/discussions/general/96417>

COURSE REQUIREMENTS:

Project Requirements	Title	Project Description	Minimum Requirements	Deliverables	Learning Emphasis
Midterm Project	Exploratory Machine Learning and Basic Prediction Task	Students shall design and implement a complete end-to-end machine learning system that addresses a real-world problem using supervised or	• Selection of a public structured dataset (e.g., CSV format) • Data cleaning (handling missing values, noise, or outliers) • Feature preparation (scaling, normalization, or encoding)	• Source code / scripts (Jupyter Notebook) • Short technical report (5–7 pages) including:	• Proper execution of the machine learning workflow • Correct application of a learning algorithm

		unsupervised learning techniques. The project requires model comparison, evaluation, and analytical reasoning, with the goal of supporting informed decision-making rather than full automation.	• Implementation of one machine learning algorithm • Regression, classification, or clustering • Use of appropriate evaluation metrics • Interpretation and discussion of results	• Dataset description • Preprocessing steps • Model implementation • Evaluation results • Brief oral presentation or demonstration	• Understanding model behavior and limitations • Emphasis on interpretation rather than mere performance scores
Final Project	Complete ML based project demonstrating complete framework	Students shall design and implement a complete end-to-end machine learning system that addresses a real-world problem using supervised or unsupervised learning techniques. The project requires model comparison, evaluation, and analytical reasoning, with the goal of supporting informed decision-making rather than full automation.	• Selection of a real-world dataset with sufficient complexity • Data preprocessing and feature engineering • Implementation of at least two machine learning models • One baseline model • One improved or alternative model • Model comparison using appropriate evaluation metrics • Discussion of limitations, assumptions, and ethical considerations • Optional: ensemble methods or hyperparameter tuning	• Complete source code with documentation • Technical report (8-12 pages) • Oral presentation and live demonstration	• Analytical reasoning and interpretation • Justification of technique selection • Ethical and responsible data use • Decision-support orientation (not full automation)
Policy Statement	Students must submit all coding tasks following proper programming standards, documentation practices, and ethical data handling guidelines. Plagiarism, data misuse, or unacknowledged code reuse shall be subject to institutional academic integrity policies.				

GRADING SYSTEM:

LABORATORY COURSES- LECTURE		PERCENTAGE
Lecture:		
➤ CLASS STANDING:	20%	
▪ Attendance		60%
▪ Oral Participation		
▪ Quizzes		
➤ MAJOR EXAMINATION:	40%	
Laboratory:		40%
➤ Performance Task	20%	
➤ Activity Output	20%	
TOTAL		100%

COURSE POLICIES:

RUBRIC FOR SCRIPT / PROGRAM / CODING STYLE					
Weight	Criteria	Excellent (4)	Very Satisfactory (3)	Satisfactory (2)	Needs Improvement (1)
A. Program Correctness & Functionality (40%)	Algorithm Implementation	Correctly implements required data mining algorithm(s) with accurate outputs	Minor errors but outputs mostly correct	Partial implementation, incorrect outputs in some cases	Algorithm not implemented or incorrect
	Data Handling	Correctly handles datasets, missing values, and edge cases	Handles most cases with minor issues	Limited handling of data results	No data validation or handling
	Output Accuracy	Results are reproducible and correct	Minor inconsistencies	Frequent incorrect results	No meaningful output
B. Code Structure & Readability (20%)	Code Organization	Well-structured, modular, clear function separation	Mostly organized, minor issues	Poor modularity	Unstructured code
	Naming Conventions	Clear, meaningful variables & function names	Mostly meaningful	Inconsistent naming	Unclear or random names

	Formatting	Proper indentation and spacing	Minor formatting issues	Inconsistent formatting	No formatting
C. Coding Style & Best Practices (15%)	Coding Standards	Follows Python / Java best practices	Minor deviations	Frequent violations	No standard followed
	Reusability	Code is reusable and extensible	Partially reusable	Limited reuse	Not reusable
D. Comments & Documentation (15%)	Inline Comments	Clear comments explaining logic	Some comments present	Few or vague comments	No comments
	Documentation	README or documentation clearly explains workflow	Basic explanation	Minimal documentation	No documentation
E. Efficiency & Tool Usage (10%)	Algorithm Efficiency	Appropriate algorithm choice and parameters	Minor inefficiencies	Inefficient implementation	Poor algorithm choice
	Tool Usage	Correct use of libraries/tools	Minor misuse	Limited usage	Incorrect usage

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